

Heat and Temperature

MCQ

1. All substances around us are made up of

- a) Cells
- b) Atoms and molecules
- c) Tissues
- d) Organs

Answer: b) Atoms and molecules

2. Atoms and molecules are always in

- a) Rest
- b) Rotational motion
- c) Vibratory motion
- d) Circular motion

Answer: c) Vibratory motion

3. Due to molecular motion, substances possess

- a) Light energy
- b) Sound energy
- c) Heat energy
- d) Electrical energy

Answer: c) Heat energy

4. Heat energy is transferred due to a difference in

- a) Mass
- b) Volume
- c) Temperature
- d) Density

Answer: c) Temperature

5. Heat flows from

- a) Colder body to hotter body
- b) Smaller body to larger body
- c) Hotter body to colder body
- d) Solid to liquid

Answer: c) Hotter body to colder body

6. Transfer of heat generally results in

- a) Increase in both temperatures
- b) Decrease in both temperatures
- c) Increase in colder body temperature
- d) No temperature change

Answer: c) Increase in colder body temperature

7. Heat is defined as

- a) Ability to do work
- b) Energy transferred due to temperature difference
- c) Measure of hotness
- d) Amount of matter

Answer: b) Energy transferred due to temperature difference

8. Heat is a

- a) Vector quantity
- b) Scalar quantity
- c) Fundamental quantity
- d) Derived force

Answer: b) Scalar quantity

9. SI unit of heat energy is

- a) Calorie
- b) Watt

- c) Joule
- d) Kelvin

Answer: c) Joule

10. Temperature indicates

- a) Quantity of heat
- b) Mass of substance
- c) Hotness or coldness
- d) Energy flow

Answer: c) Hotness or coldness

11. Heat is the cause and temperature is the

- a) Effect
- b) Energy
- c) Unit
- d) Motion

Answer: a) Effect

12. An object is called hot when its temperature is

- a) Equal to body temperature
- b) Less than body temperature
- c) Higher than body temperature
- d) Zero

Answer: c) Higher than body temperature

13. Hot objects have atoms that move

- a) Slowly
- b) Randomly
- c) Very fast
- d) Not at all

Answer: c) Very fast

14. An object feels cold when its temperature is

- a) Higher than body temperature
- b) Equal to body temperature
- c) Lower than body temperature
- d) Very high

Answer: c) Lower than body temperature

15. Cold objects have atoms that move

- a) Very fast
- b) Slowly
- c) Violently
- d) Uniformly fast

Answer: b) Slowly

16. Common units used to measure temperature are

- a) Joule, Watt, Kelvin
- b) Celsius, Fahrenheit, Kelvin
- c) Meter, Kelvin, Joule
- d) Degree, Watt, Calorie

Answer: b) Celsius, Fahrenheit, Kelvin

17. Celsius scale is also called

- a) Absolute scale
- b) Kelvin scale
- c) Centigrade scale
- d) British scale

Answer: c) Centigrade scale

18. SI unit of temperature is

- a) Degree Celsius
- b) Degree Fahrenheit

c) Kelvin

d) Joule

Answer: c) Kelvin

19. 100°C is read as

a) Hundred Celsius

b) Hundred degree Celsius

c) Hundred Kelvin

d) Hundred Fahrenheit

Answer: b) Hundred degree Celsius

20. Higher temperature means molecules move

a) Slowly

b) Not at all

c) Rapidly

d) Uniformly

Answer: c) Rapidly

21. When temperature is zero, molecular motion

a) Increases

b) Remains constant

c) Stops

d) Becomes random

Answer: c) Stops

22. On heating, substances generally

a) Contract

b) Expand

c) Break

d) Melt only

Answer: b) Expand

23. On cooling, substances generally

- a) Expand
- b) Contract
- c) Evaporate
- d) Burn

Answer: b) Contract

24. Instrument used to measure temperature is

- a) Barometer
- b) Thermometer
- c) Hygrometer
- d) Ammeter

Answer: b) Thermometer

25. Thermometers work on the principle of

- a) Reflection
- b) Radiation
- c) Expansion and contraction
- d) Conduction

Answer: c) Expansion and contraction

26. Common liquids used in thermometers are

- a) Water and oil
- b) Mercury and alcohol
- c) Acid and base
- d) Milk and water

Answer: b) Mercury and alcohol

27. Mercury is used in thermometers because it

- a) Evaporates easily
- b) Has uniform expansion

- c) Is transparent
- d) Freezes quickly

Answer: b) Has uniform expansion

28. Mercury does not stick to

- a) Plastic
- b) Metal
- c) Glass
- d) Rubber

Answer: c) Glass

29. Boiling point of mercury is about

- a) 100°C
- b) 0°C
- c) 357°C
- d) -39°C

Answer: c) 357°C

30. Alcohol is suitable for measuring very low temperatures because

- a) It boils fast
- b) It is colourless
- c) It has very low freezing point
- d) It is heavy

Answer: c) It has very low freezing point

31. Clinical thermometer is used to measure

- a) Room temperature
- b) Water temperature
- c) Body temperature
- d) Soil temperature

Answer: c) Body temperature

32. Mercury in a clinical thermometer acts as

- a) Scale
- b) Sensor
- c) Container
- d) Insulator

Answer: b) Sensor

33. The kink in a clinical thermometer helps to

- a) Expand mercury
- b) Stop mercury from falling back
- c) Increase accuracy
- d) Cool mercury

Answer: b) Stop mercury from falling back

34. Normal human body temperature is

- a) 35°C
- b) 36°C
- c) 37°C
- d) 38°C

Answer: c) 37°C

35. Body temperature is usually measured in

- a) °C
- b) K
- c) °F
- d) J

Answer: c) °F

36. Range of a clinical thermometer is

- a) 0°C to 100°C
- b) 35°C to 42°C

c) -10°C to 110°C

d) 32°F to 212°F

Answer: b) 35°C to 42°C

37. Laboratory thermometers are used when temperature

a) Is less than 0°C only

b) Is body temperature

c) Is outside clinical thermometer range

d) Is constant

Answer: c) Is outside clinical thermometer range

38. Laboratory thermometers do not have

a) Bulb

b) Mercury

c) Scale

d) Kink

Answer: d) Kink

39. Temperature range of laboratory thermometer is

a) 35°C to 42°C

b) 0°C to 100°C

c) -10°C to 110°C

d) 32°F to 212°F

Answer: c) -10°C to 110°C

40. While using lab thermometer, it should be

a) Tilted

b) Shaken

41. Digital thermometers work on the principle of change in

- a) Volume
- b) Resistance
- c) Density
- d) Pressure

Answer: b) Resistance

42. Digital thermometers are preferred because they are

- a) Slower
- b) Less accurate
- c) Toxic
- d) Fast and accurate

Answer: d) Fast and accurate

43. Infrared thermometers measure temperature by detecting

- a) Sound waves
- b) Light rays
- c) Infrared radiation
- d) Electric current

Answer: c) Infrared radiation

44. Infrared thermometers are useful because they

- a) Touch the object
- b) Work only in labs
- c) Measure from a distance
- d) Use mercury

Answer: c) Measure from a distance

45. Kelvin scale starts from

- a) 0°C
- b) 32°F

- c) Absolute zero
- d) Freezing point

Answer: c) Absolute zero

46. Freezing point of water in Celsius scale is

- a) 32°C
- b) 100°C
- c) 0°C
- d) -10°C

Answer: c) 0°C

47. Boiling point of water in Fahrenheit scale is

- a) 100°F
- b) 212°F
- c) 32°F
- d) 373°F

Answer: b) 212°F

48. Temperature scales are based on

- a) Human body temperature
- b) Sun temperature
- c) Freezing and boiling points of water
- d) Room temperature

Answer: c) Freezing and boiling points of water

49. Heat depends on

- a) Temperature only
- b) Mass and temperature
- c) Shape of object
- d) Colour of object

Answer: b) Mass and temperature

50. Temperature is independent of

- a) Hotness
- b) Coldness
- c) Quantity of substance
- d) Molecular motion

Answer: c) Quantity of substance